

Big Data Analytics in Cloud Computing

Slot-C

CSA1578

Cloud Computing and Big Data Analytics

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Objective

- Understand the concept of Big Data and Cloud Computing.
- Learn how cloud platforms process and analyze massive datasets.
- Study cloud-based big data architectures and analytics platforms.
- Explore business benefits and real-world applications.
- Understand future trends and challenges.

Core Technical Explanation

What is Big Data?

- Big Data refers to extremely large and complex datasets that cannot be processed efficiently using traditional database systems.

Characteristics (5 Vs)

- **Volume:** Huge amounts of data (TBs, PBs).
- **Velocity:** Fast generation and processing of data.
- **Variety:** Structured, semi-structured, and unstructured data.
- **Veracity:** Data quality and reliability.
- **Value:** Useful insights obtained from data.

What is Cloud Computing?

- Cloud Computing provides computing resources such as servers, storage, databases, networking, and software over the Internet on a pay-as-you-use basis.

Why Use Cloud for Big Data?

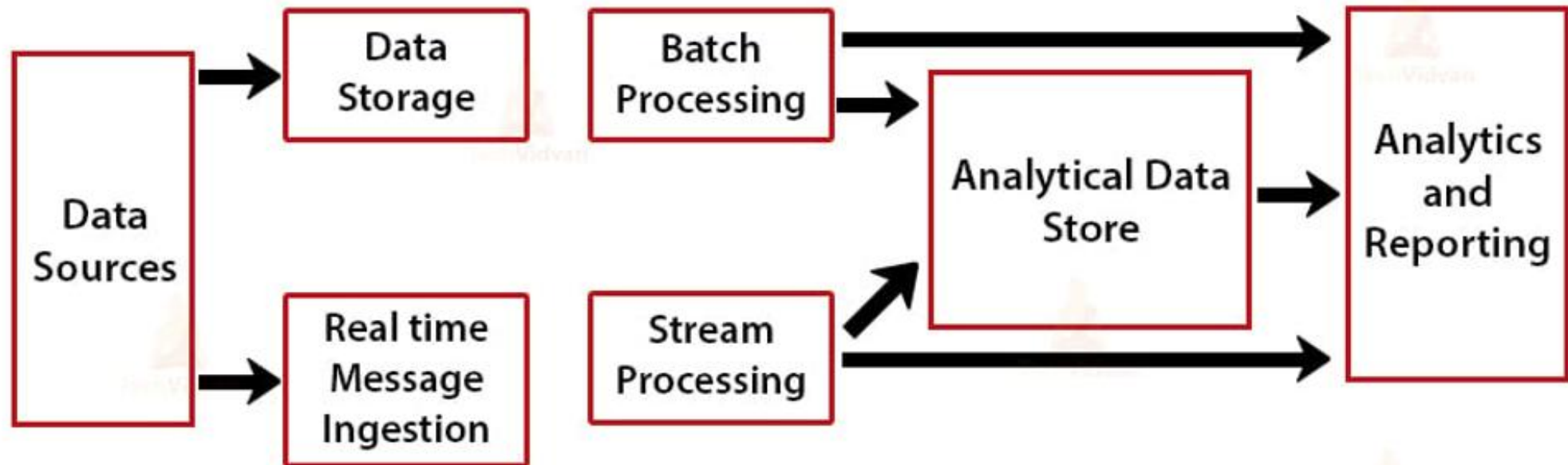
- Unlimited storage capacity
- High-performance computing
- Real-time analytics
- Cost-effective infrastructure
- Automatic scalability
- Global accessibility

Technologies Used

- **Apache Hadoop** – Distributed storage and processing.
- **Apache Spark** – Fast in-memory data processing.
- **Kafka** – Real-time data streaming.
- **Amazon S3** – Cloud object storage.
- **Google BigQuery** – Serverless data warehouse.
- **Azure Synapse Analytics** – Enterprise analytics platform.

Architecture Diagram

Big Data Architecture



Real-World Use Case

Netflix Recommendation System

Problem

- Netflix receives billions of user interactions every day, making traditional data processing inefficient.

Cloud-Based Solution

- User activity is stored in cloud storage.
- Apache Spark processes viewing history.
- Machine Learning predicts user preferences.
- Recommendations are displayed in real time.

Benefits

- Personalized recommendations
- Increased customer satisfaction
- Reduced infrastructure costs
- Faster content delivery

Conclusion and Key Takeaway

Conclusion

- Big Data Analytics combined with Cloud Computing enables organizations to process, store, and analyze massive datasets efficiently. Cloud platforms provide scalability, flexibility, and cost-effective infrastructure, helping businesses make faster and smarter decisions.

Key Takeaways

- Cloud enables scalable Big Data processing.
- Distributed computing improves performance.
- Real-time analytics supports quick decision-making.
- Machine Learning enhances predictive analytics.
- Cloud analytics reduces infrastructure costs.
- Big Data drives innovation across industries.

Limitations and Future Scope

Limitations

- Data security and privacy concerns
- High network dependency
- Vendor lock-in
- Data migration complexity
- Compliance and regulatory issues
- Large data transfer costs

Future Scope

- AI-powered cloud analytics
- Edge computing integration
- Serverless data analytics
- Quantum computing for faster processing
- Green and sustainable cloud computing
- Real-time predictive analytics
- Automated data governance
- Integration with Internet of Things (IoT)

*Thank
you!*

GEO TAG:

